# Artificial Intelligence

## Deep learning 5.5 - *Generative models*

*Master in Artificial Intelligence & Computational engineering*

Ana C. Murillo

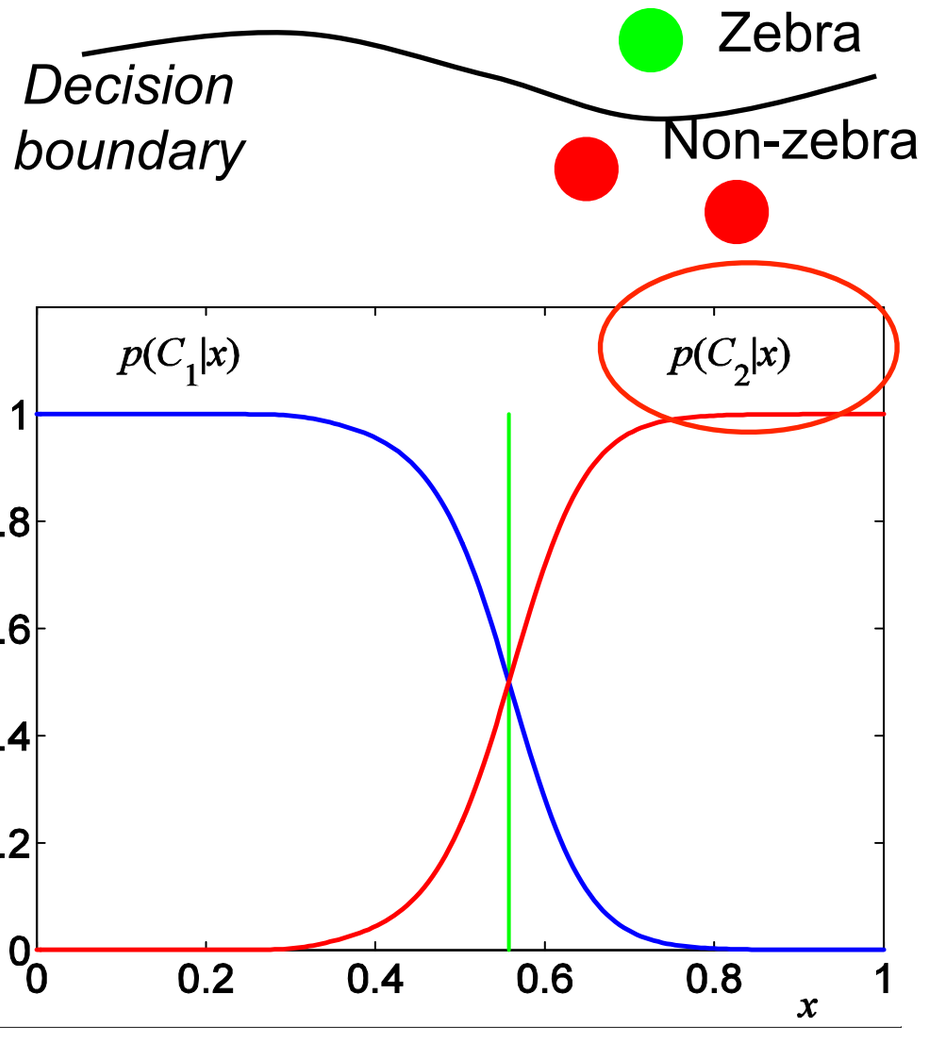
Universidad Zaragoza
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# This block

- **Generative models**:
  - Autoencoder
  - Generative Adversarial Networks
  - Diffusion

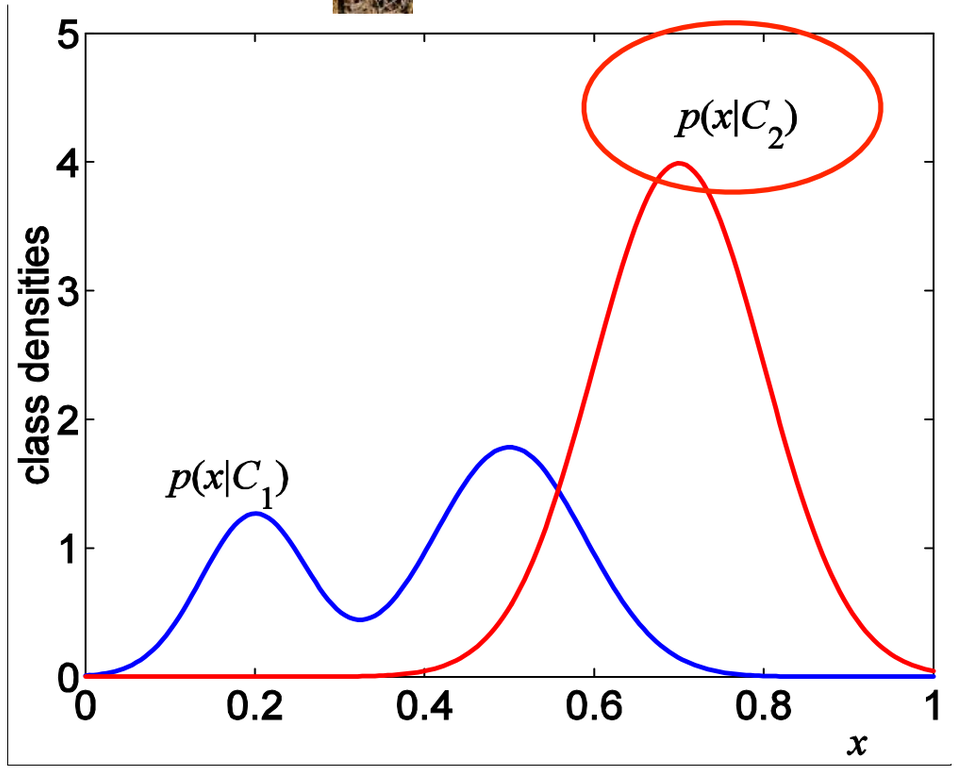
# Deep Learning && Generative

- **Discriminative** models:
  - posterior $$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}$$



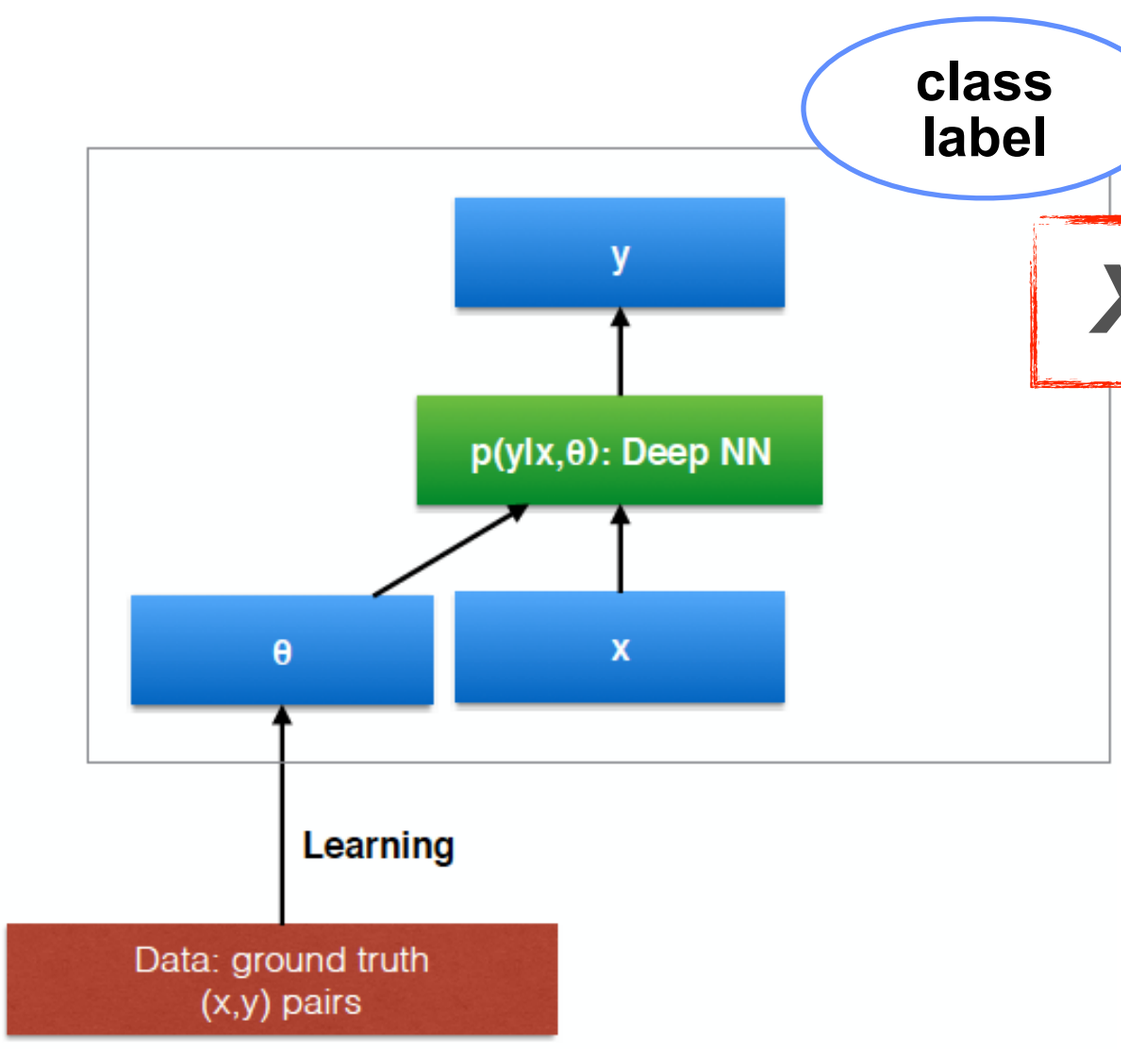

- **Generative** models:
  - likelihood and priors



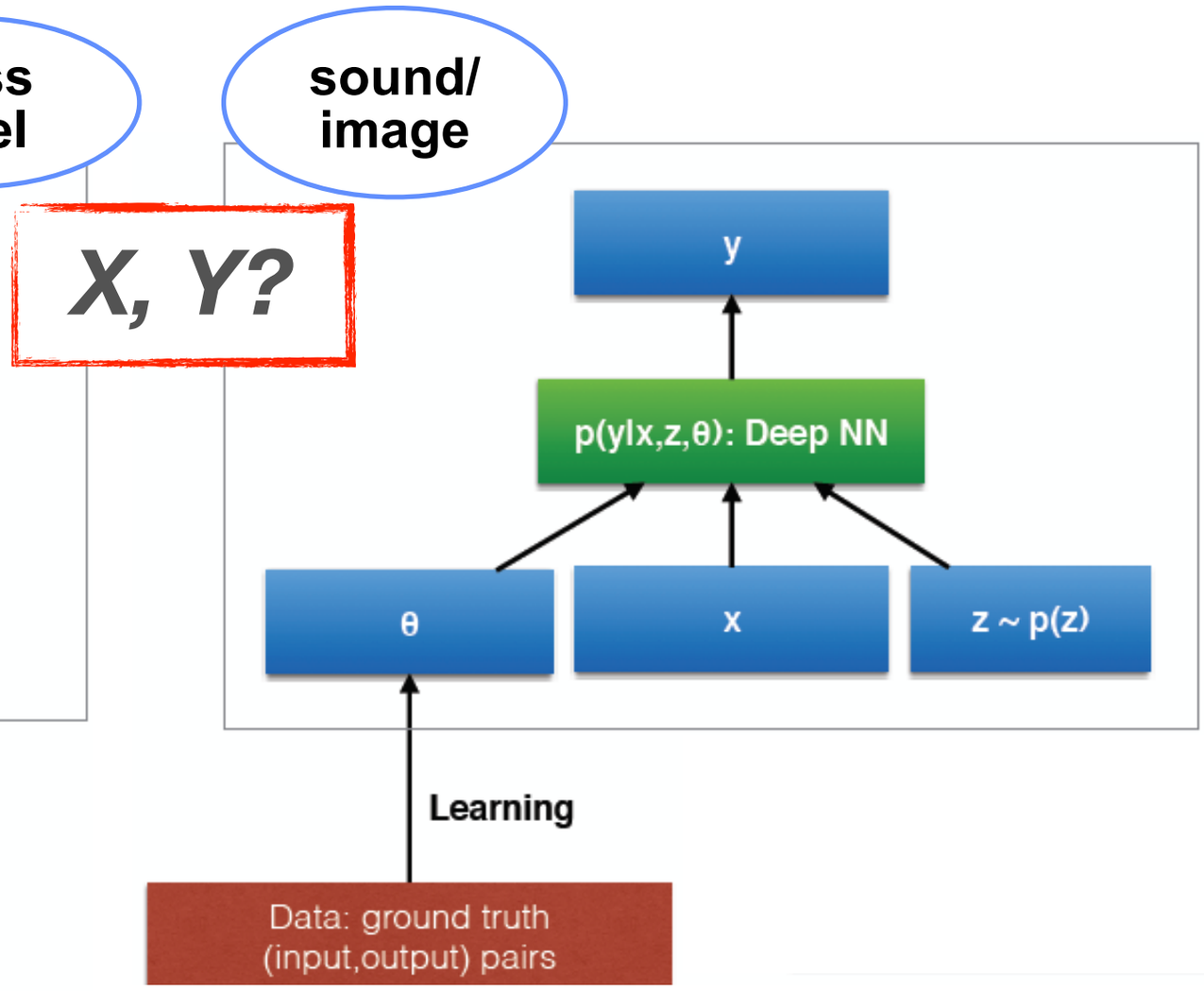


# Deep Learning && Generative

- **Discriminative** models:




- **Generative** models:




X, Y?

*Durk Kingma*
*Berkeley Lecture slides - 2016*

# *Foundation* models & image generation

## LLM + diffusion model. **Images from "*prompts*"**

Two self-supervised generative objectives

**Language model**
Predicts the next token (discrete)

**Diffusion model**
Denoises a latent step by step (continuous)

**LLM conditions diffusion**

Separate models

Text prompt

LLM / text encoder → embeddings

Diffusion transformer (denoiser)

VAE decoder → image

*e.g. Stable Diffusion 3, Imagen, Qwen-Image*

**Post-training.** Preference optimization and RL now also tune image generators, like for text.

# Deep Learning && Generative

**Assumption**: data comes from a unknown data distribution

**Goal**: train networks to learn this distribution from its samples

… to generate new data by sampling from the learned model

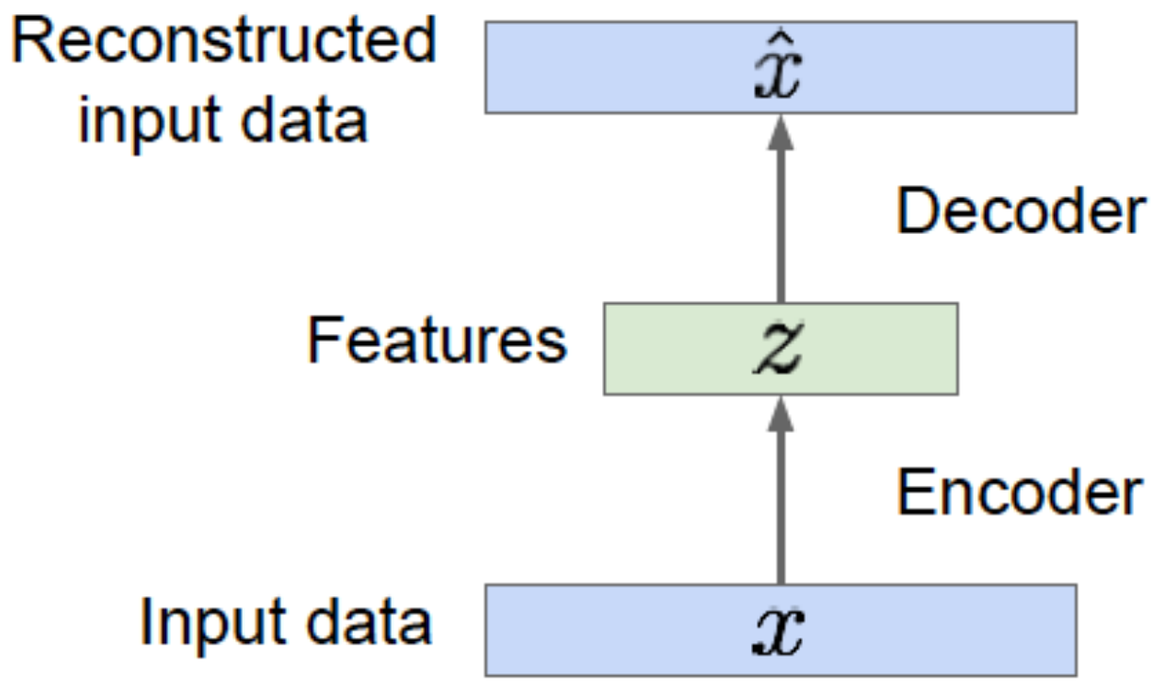
Many **generative models**, with different types of supervision:

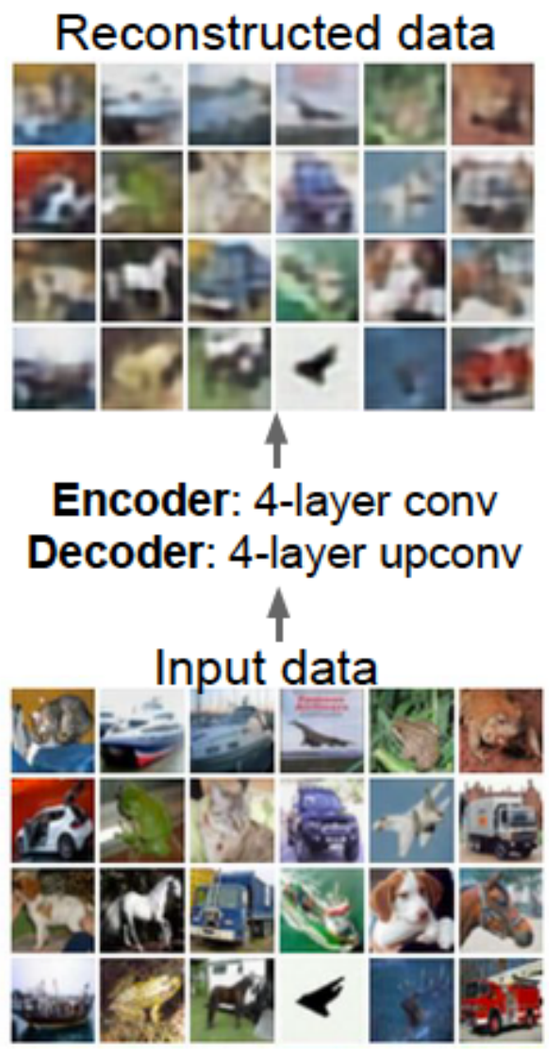
- Pixel RNN,
- Autoencoders,
- Generative Adversarial Networks (GAN),
- Diffusion-based,
- etc …

# Deep Learning && Generative: **Autoencoder**

- **Auto encoders**
  - Similar *encoder-decoder* architecture than segmentation BUT unsupervised
  - Learn features that can reconstruct original images

***How to learn this?***

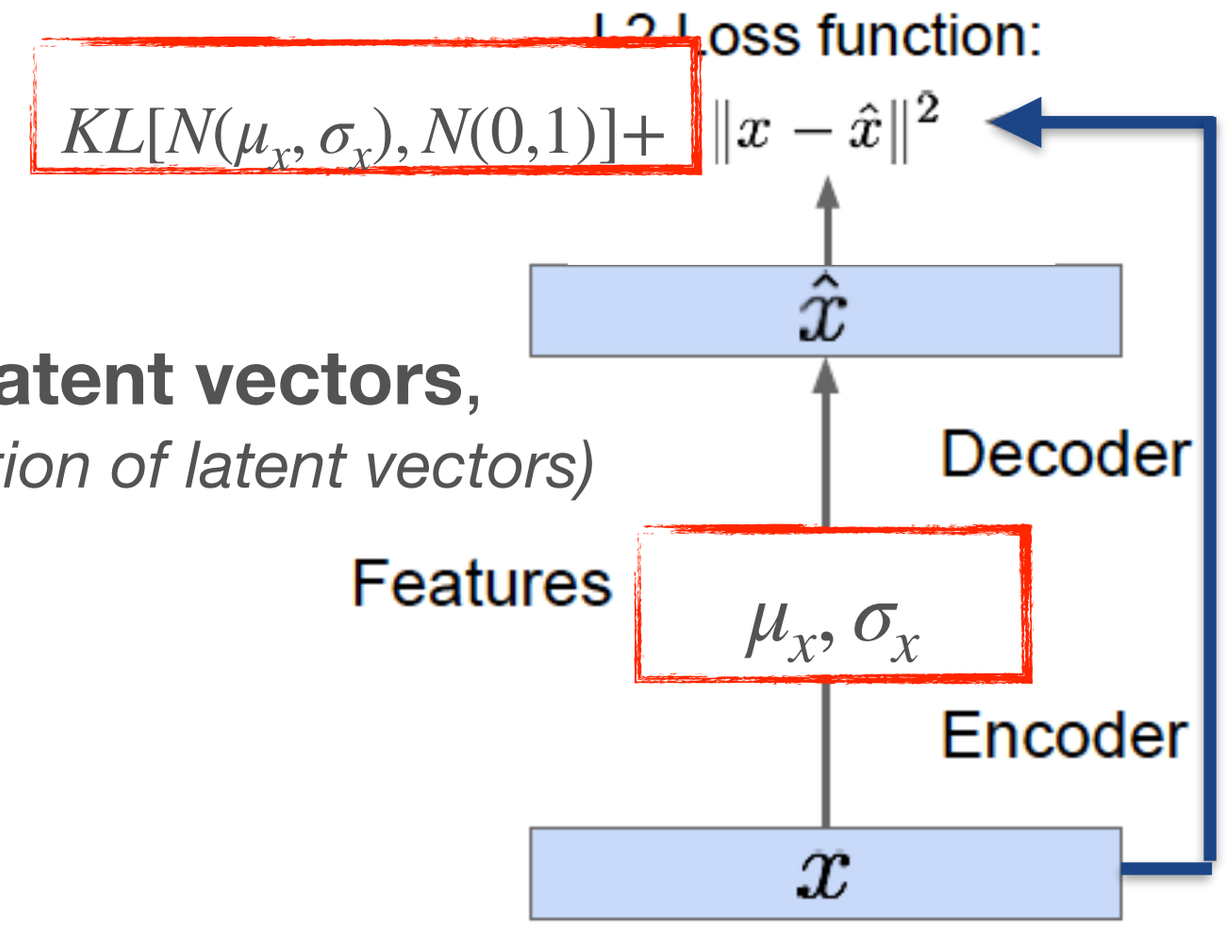



*Adapted from Li, Johnson, Yeung. http://cs231n.stanford.edu 2017*

# Deep Learning && Generative

- Auto-encoder:
  - **deterministic** mapping: input —> **latent vector**
  - only the outputs of the encoder can be decoded
  - Decoding a *random* latent might give garbage




- **Variational Auto encoders**
  - mapping to a **distribution of latent vectors**,
    *(i.e., mean & covariance of distribution of latent vectors)*
  - Train with reconstruction loss
    + **regularizing term**
  - probabilistic encoding enforces **structured latents**
    —> better results for decoding random latent inputs

# Deep Learning && Generative - **GAN**

- **Goal**: realistic results from image decoders
- Can I **automatically evaluate** how “realistic” my decoder is?

**Generative Adversarial Networks (GAN)**

# Deep Learning && Generative

- **Generative Adversarial Networks (GAN)**

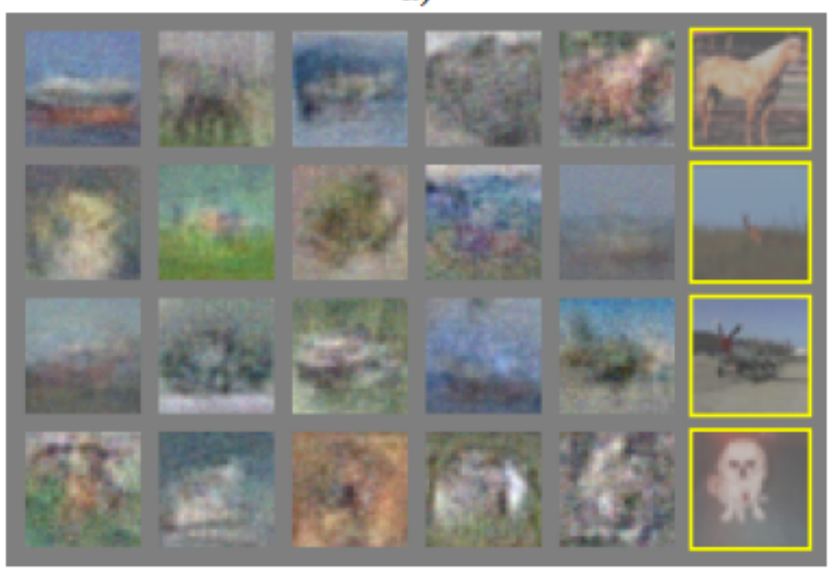
Train 2 networks simultaneously to generate realistic data



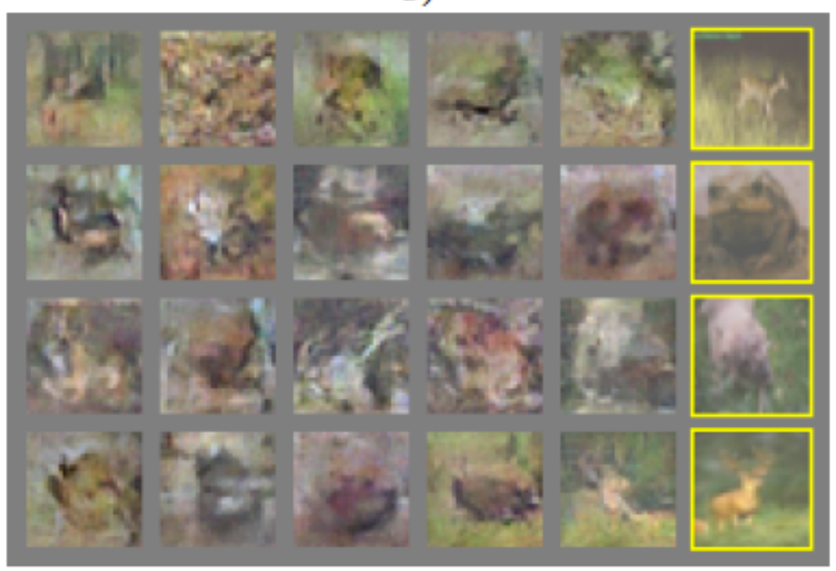
a)



b)



c)



d)

*samples from the generator networks*

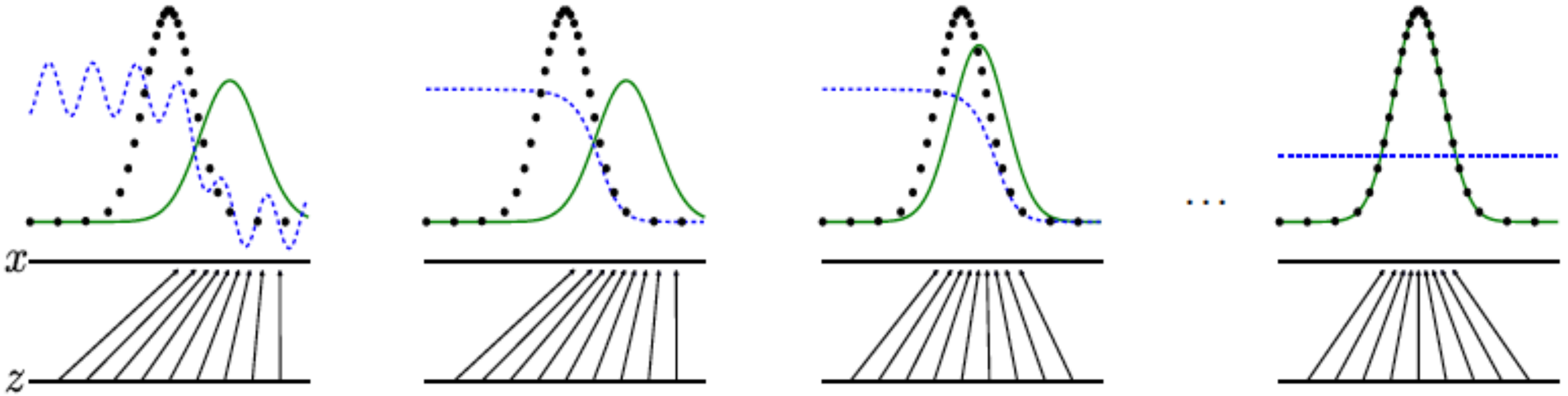
*Goodfellow, Ian, et al. "Generative adversarial nets." NIPS 2014.*

# Deep Learning && Generative

- **Generative Adversarial Networks (GAN)**

  Train **2 networks, *G* and *D***: *How?*

  - **Generative** model (captures data distribution) —> maximize probability of D making a mistake
  - **Discriminative** model—> discriminate “actual” data from “generated” data by G, i.e., probability of a sample coming from the training data or G.



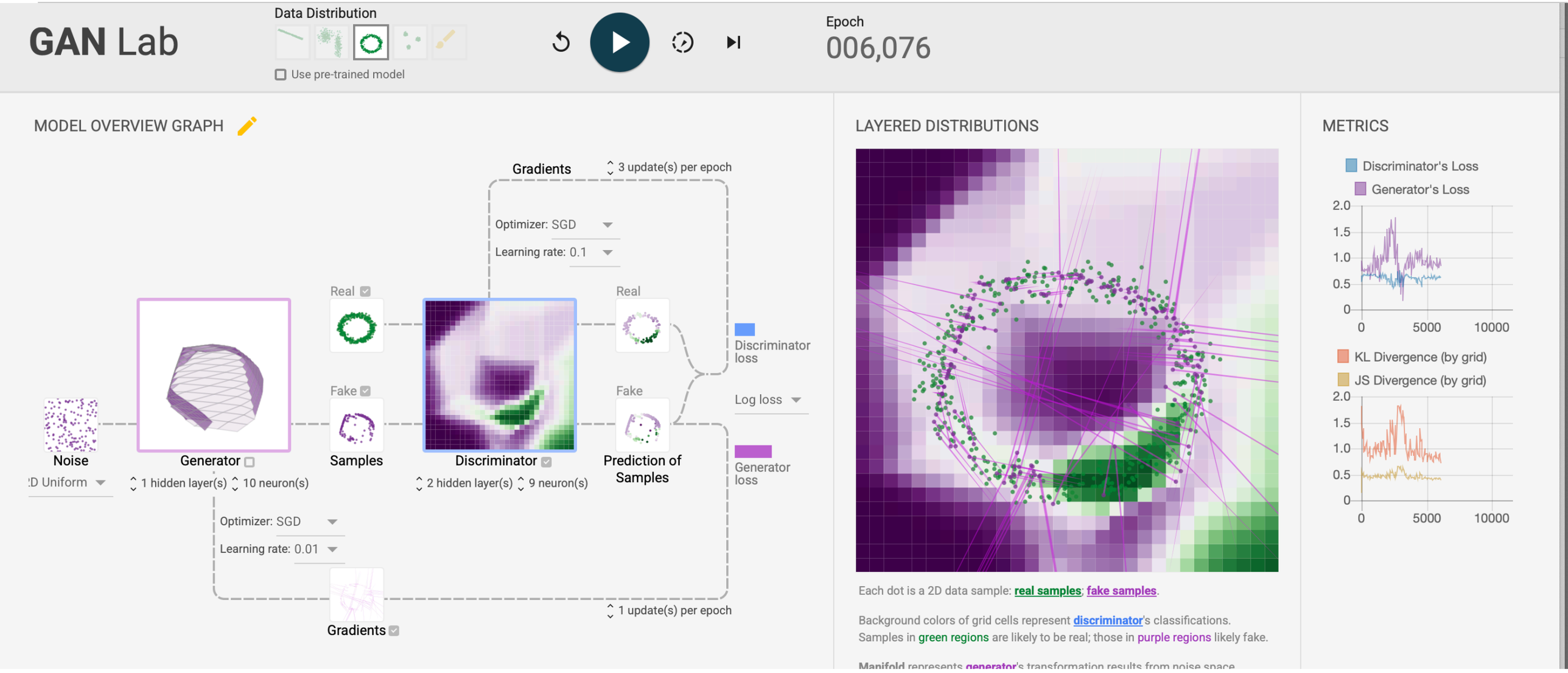

DEMO-web: https://cs.stanford.edu/people/karpathy/gan

*Goodfellow, Ian, et al. "Generative adversarial nets” NIPS 2014.*

# Deep Learning && Generative

- **Generative Adversarial Networks (GAN)**

  Train **2 networks, *G* and *D***: *How?*



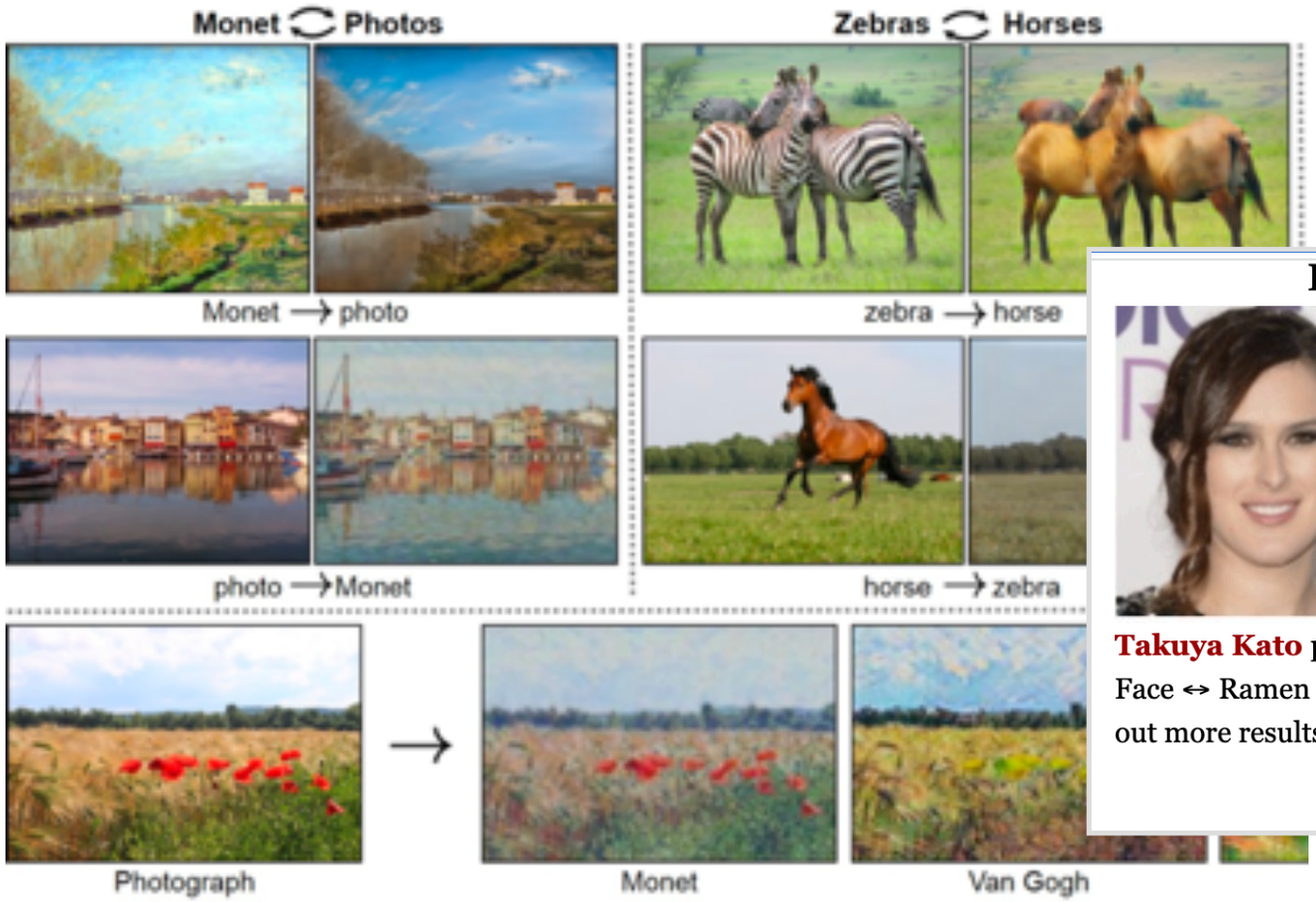

https://poloclub.github.io/ganlab/

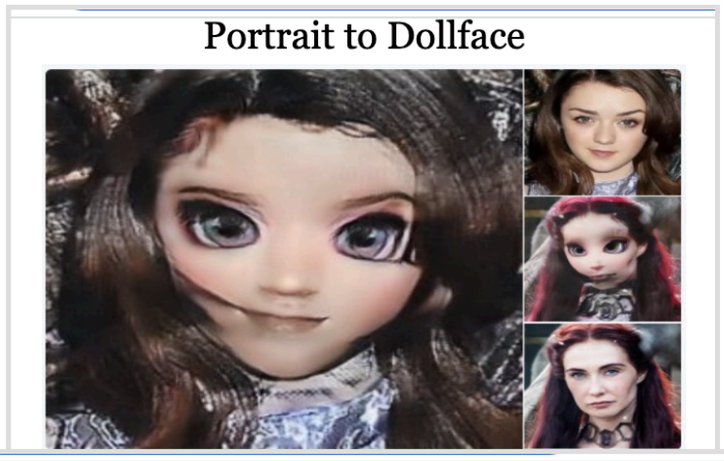
# Deep Learning && Generative

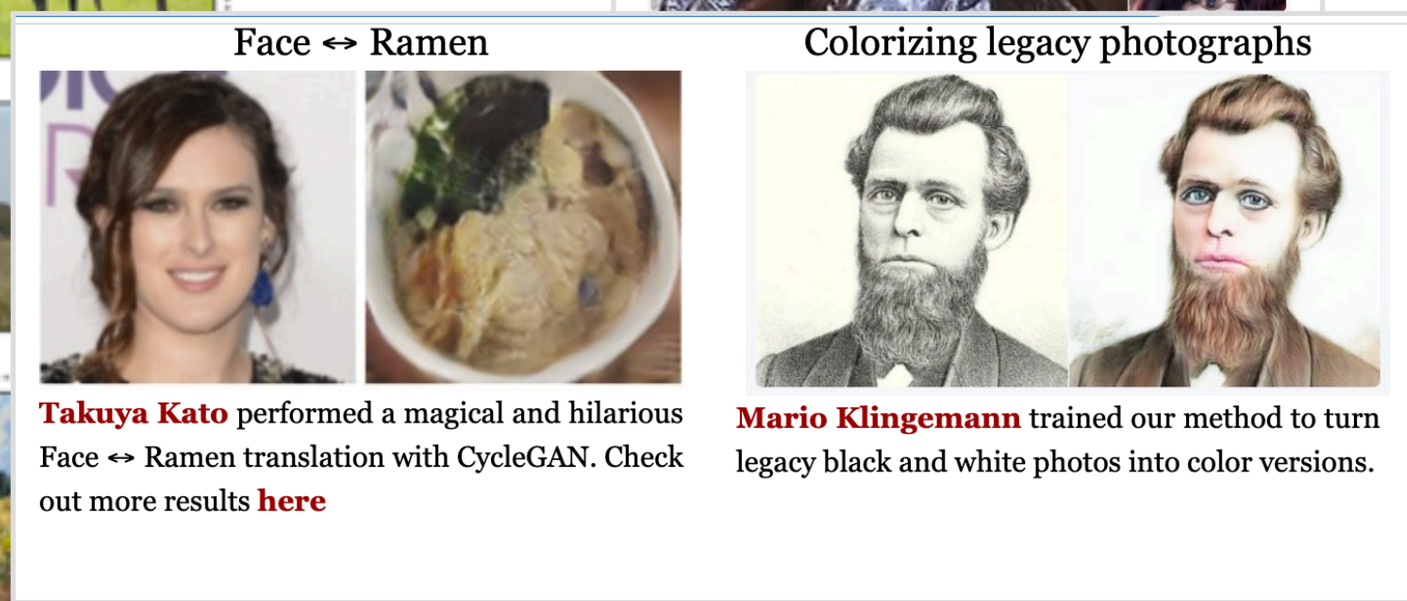
- **Generative Adversarial Networks** (GAN)
  - Supervised … but “weakly”
  - Known to be **very *hard (tricky)* to train**
    - Generator stops learning (e.g. If the discriminator is too good, gradients to the generator can vanish)
    - Generator might slip up and get caught in a “mode collapse” (e.g. finds a few outputs that perform well at fooling the discriminator, and outputs only those for all inputs)
    - Training procedure issues (The vanilla GAN setup doesn’t converge, wild oscillations during training)
  - Many strategies proposed to improve these issues

# Deep Learning && Generative

- CycleGAN: Gan losses beyond raw image generation… used to ensure “quality” and “realism” in different tasks
  https://junyanz.github.io/CycleGAN/




*J. Zhu, T. Park, P. Isola. A. Efros,*
*"Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks," ICCV 2017.*

# Deep Learning && Generative

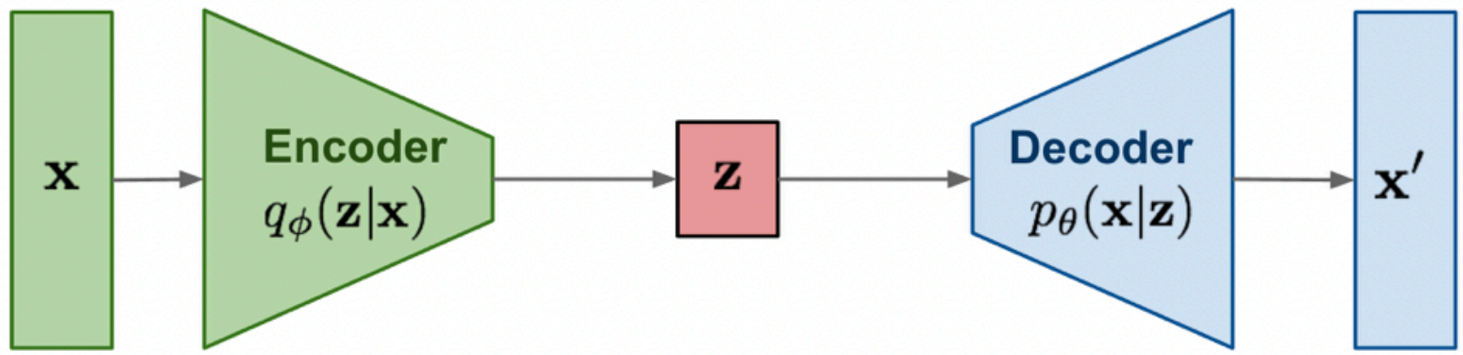


*Gatys, Leon A., Alexander S. Ecker, and Matthias Bethge. "Image style transfer using convolutional neural networks." CVPR. 2016.*
*Johnson, Justin, Alexandre Alahi, and Li Fei-Fei. "Perceptual losses for real-time style transfer and super-resolution" ECCV 2016.*

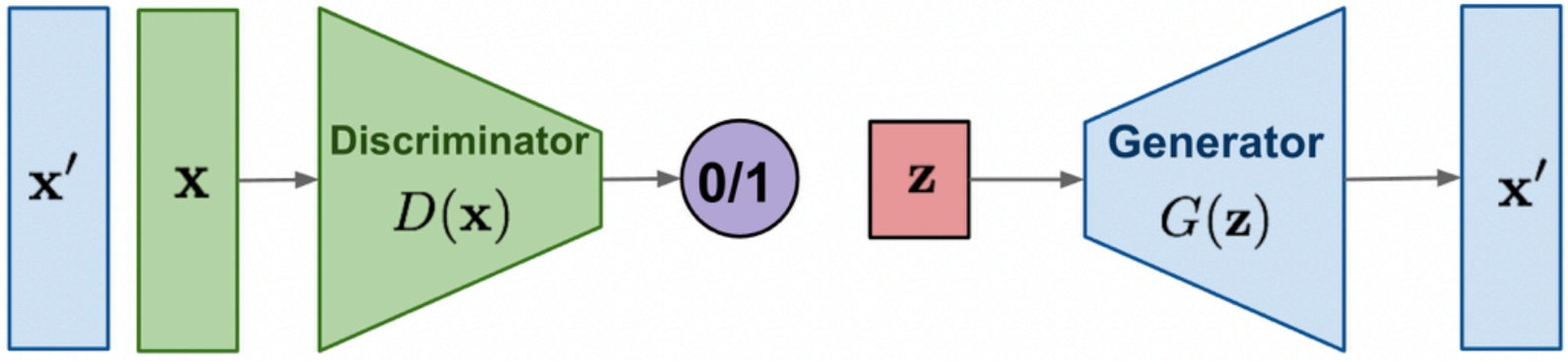
- DEMO-web: https://tenso.rs/demos/fast-neural-style/
- A lot of interactive demos:
  https://vcai.mpi-inf.mpg.de/projects/DragGAN/
  https://www.nvidia.com/en-us/research/ai-playground/
- Of course … deep-fakes: https://deepfakedetectionchallenge.ai/
- Although lately GANs are losing impact regarding image generation with respect to new text2im approaches using **diffusion models** …

# Deep Learning && Generative

**VAE:** maximize variational lower bound



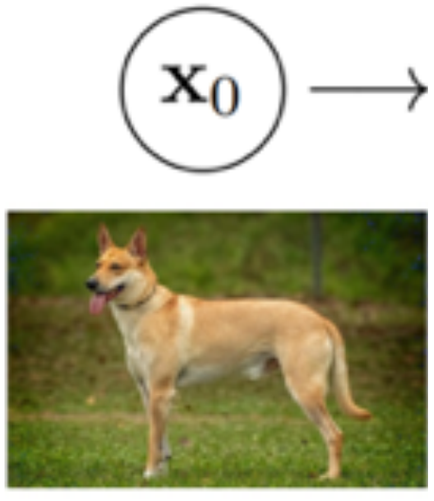

**GAN:** Adversarial training



# Deep Learning && Generative - **Diffusion**

**Diffusion models - Forward diffusion process:**

- Gradually add very small amounts of Gaussian noise to input data x_0
- Repeat this process for T steps until you get something resembling pure noise

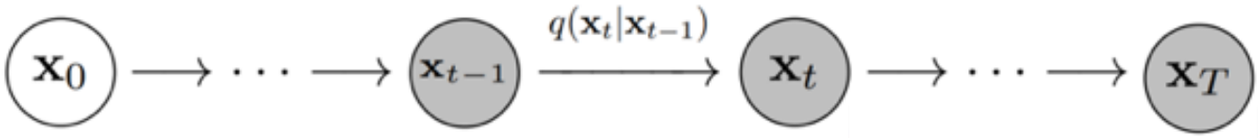


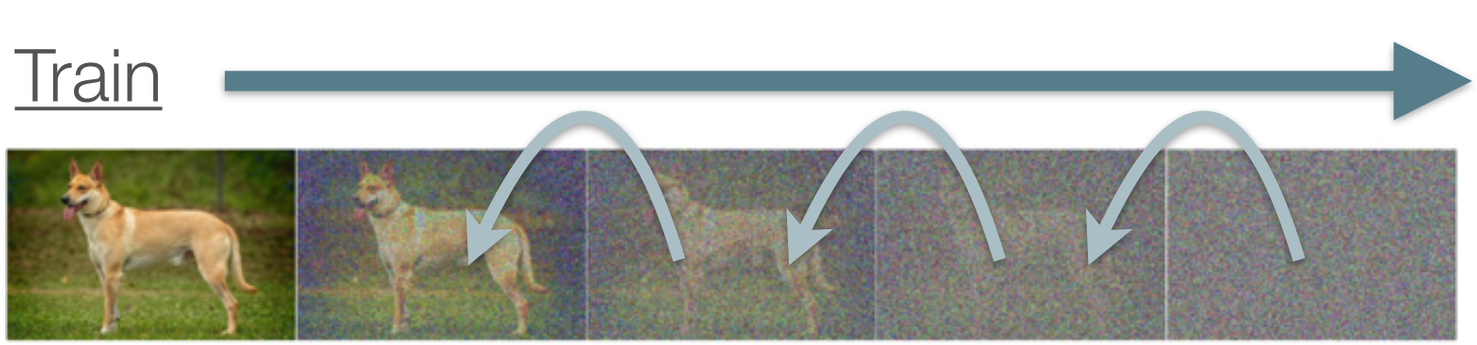

*Ho, Jonathan, Ajay Jain, and Pieter Abbeel.*
*“Denoising diffusion probabilistic models." NeurIPS 2020.*

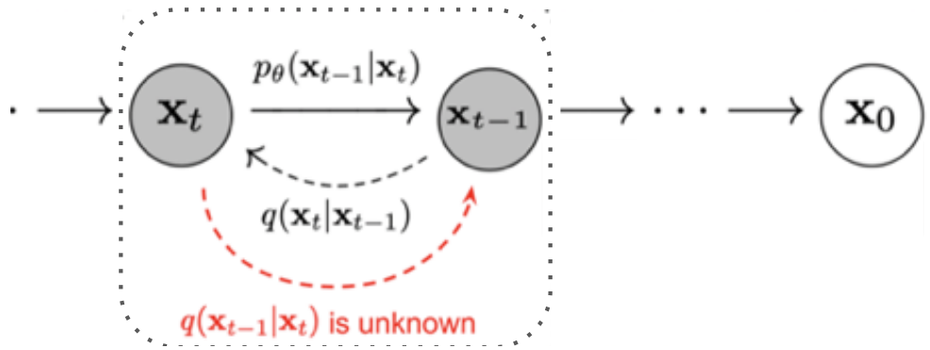
# Deep Learning && Generative

## Diffusion models - reverse step

- Learn reverse denoising process to iteratively undo the forward diffusion process
- The reverse process seems to be generating new data from random noise



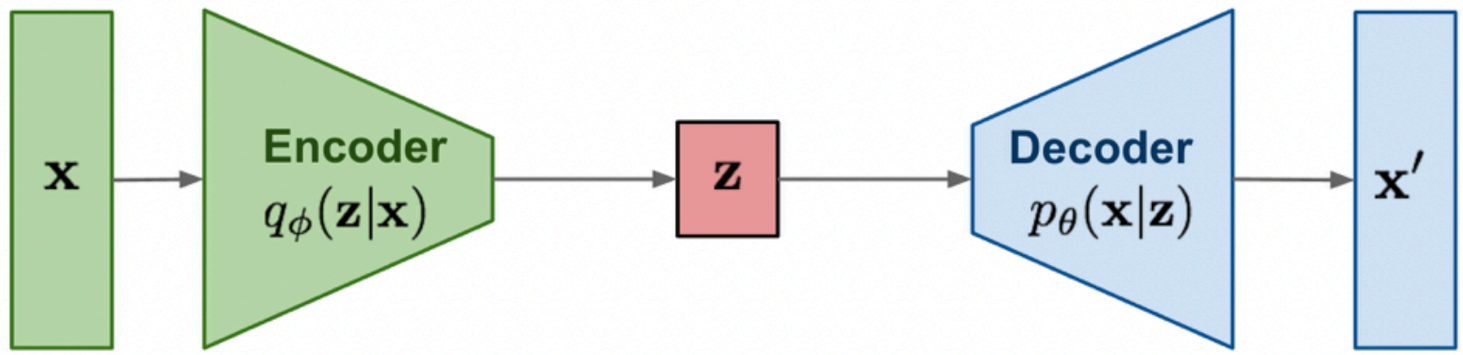



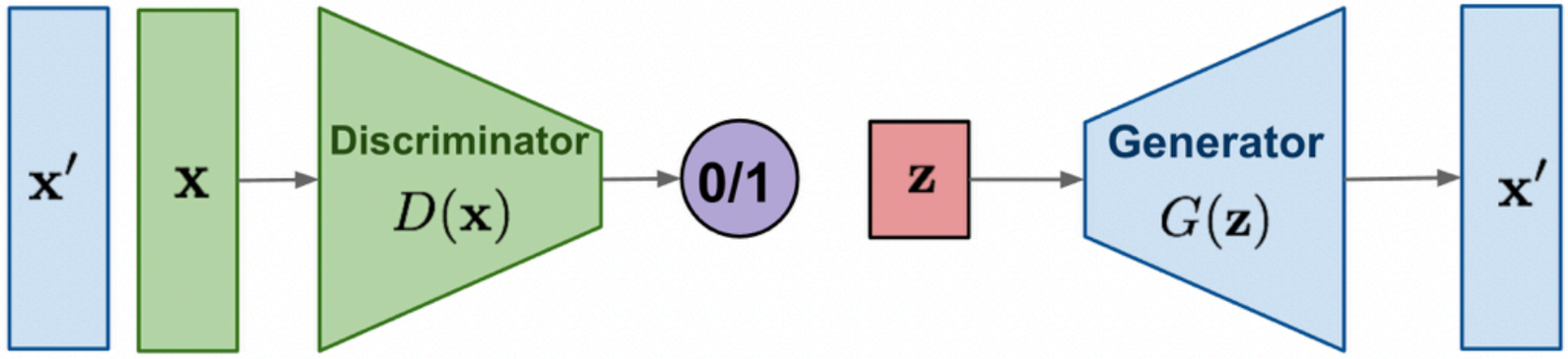

*Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." NeurIPS 2020.*

# Deep Learning && Generative

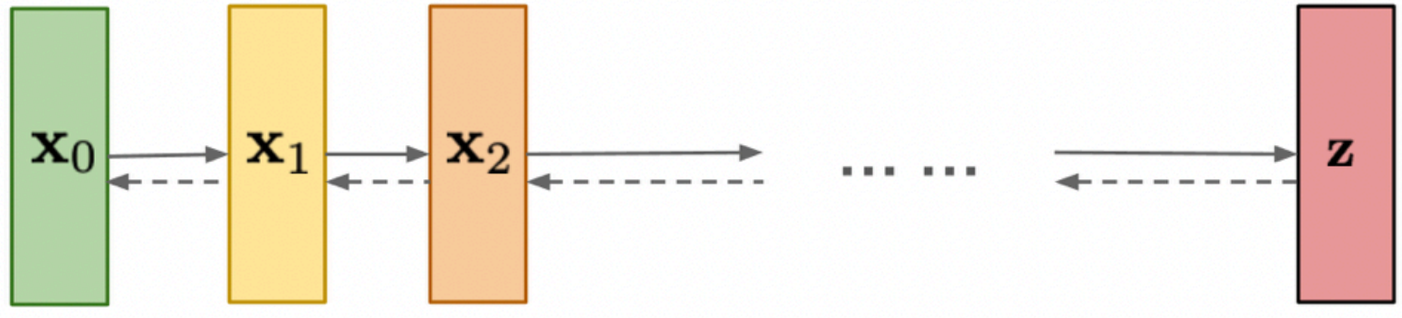
**VAE:** maximize variational lower bound




**GAN:** Adversarial training




**Diffusion models:** Gradually add Gaussian noise and then reverse




*CS 198-126. Deep learning for visual data. UC Berkeley*

# Deep Learning && Generative

**Diffusion models + ATTENTION + …**

- Text to image: Dall-E (openAI), Imagen 3 (Google)
- Image to image: https://imagic-editing.github.io/
- Text to video: https://imagen.research.google/video/
- Many more …

# Deep Learning && Generative - *Diffusion*

## LLM + diffusion model. **Images from "*prompts*"**

Two self-supervised generative objectives

**Language model**
Predicts the next token (discrete)

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Denoises a latent step by step (continuous)

**LLM conditions diffusion**

Separate models

Text prompt

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Diffusion transformer (denoiser)

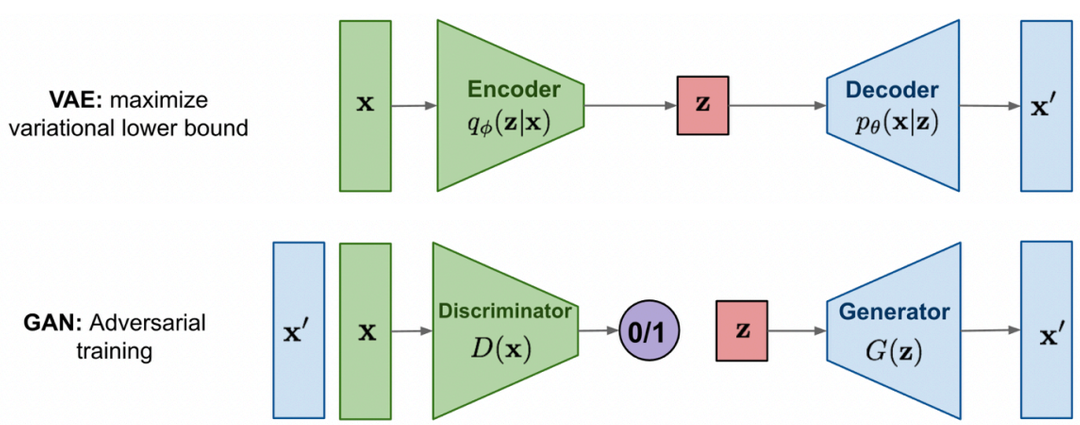
VAE decoder → image

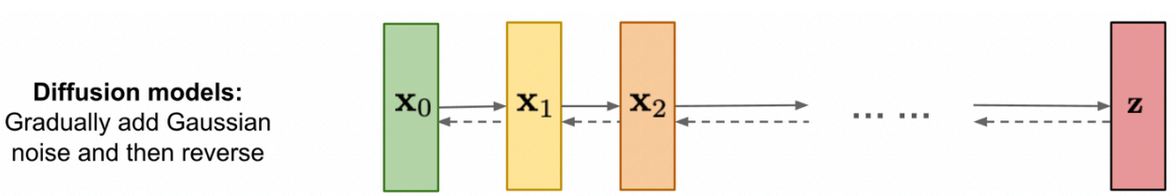
*e.g. Stable Diffusion 3, Imagen, Qwen-Image*

**Post-training.** Preference optimization and RL now also tune image generators, like for text.

# Deep Learning && Generative

| Feature | Autoencoder (AE) | VAE | GAN (Adversarial) | Diffusion |
|---|---|---|---|---|
| *Core Goal/Basis* | | ?? | | |
| *Training Stability* | | Stable? | | |
| *Inference Speed* | | Fast? | | |
| *Image Quality* | | Good? | | |
| *Latent Space* | | ?? | | |
| *Good use cases* | | | | |




*CS 198-126. Deep learning for visual data. UC Berkeley*

# Main resources for the materials in this class

- CS294-129 Designing, Visualizing and Understanding Deep Neural Networks. UC Berkeley.

- CS 198-126. Deep learning for visual data. UC Berkeley.

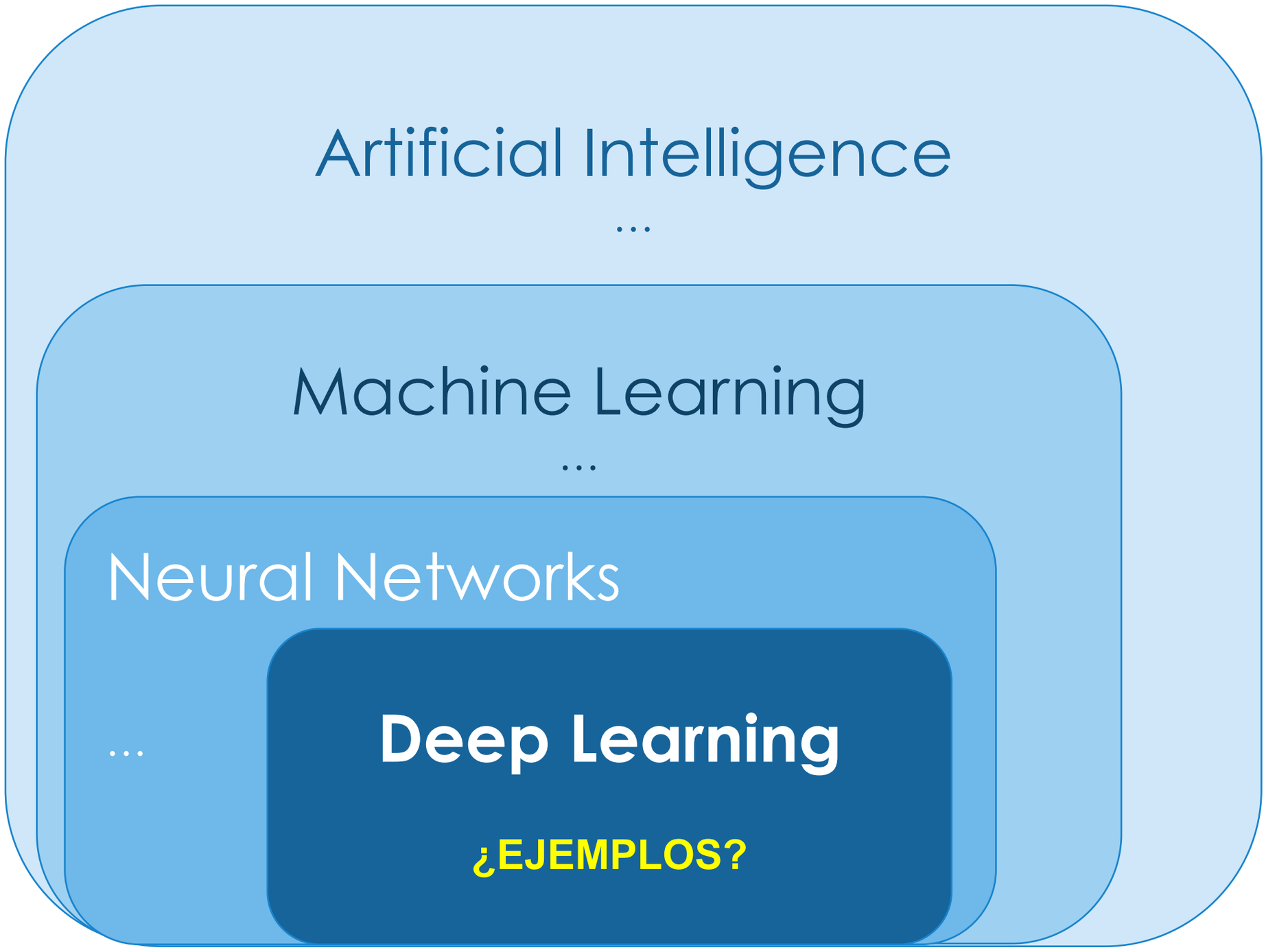
- Stanford classes on deep learning for Computer Vision http://cs231n.stanford.edu

- *Computer Vision: Algorithms and Applications*. 2nd Edition. Richard Szeliski. https://szeliski.org/Book/

# Thanks for attending! This was the last “lecture”

- **Tomorrow, last day, LAB 5**
  - If possible review the prior work to be prepared
  - It’s short, to submit right after the session or next day. We’ll evaluate it on Thursday with an interview.
- **Final Exam:** short questions (everyone, around 1h) + Global exam about labs (if not submitted in the continuous evaluation)
- Remember to submit Lab4

# Note that … AI & ML



# Note that … AI & ML

